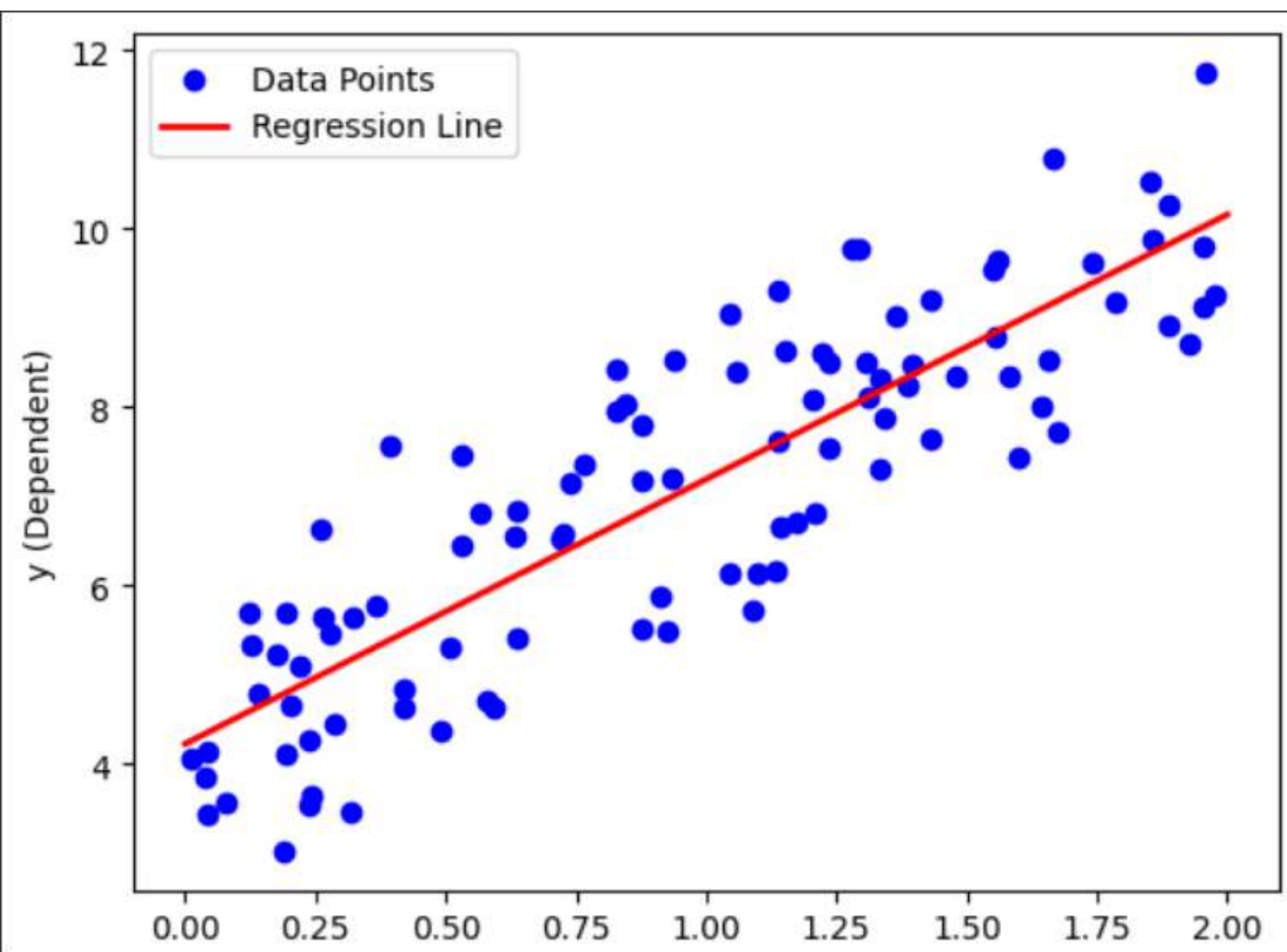
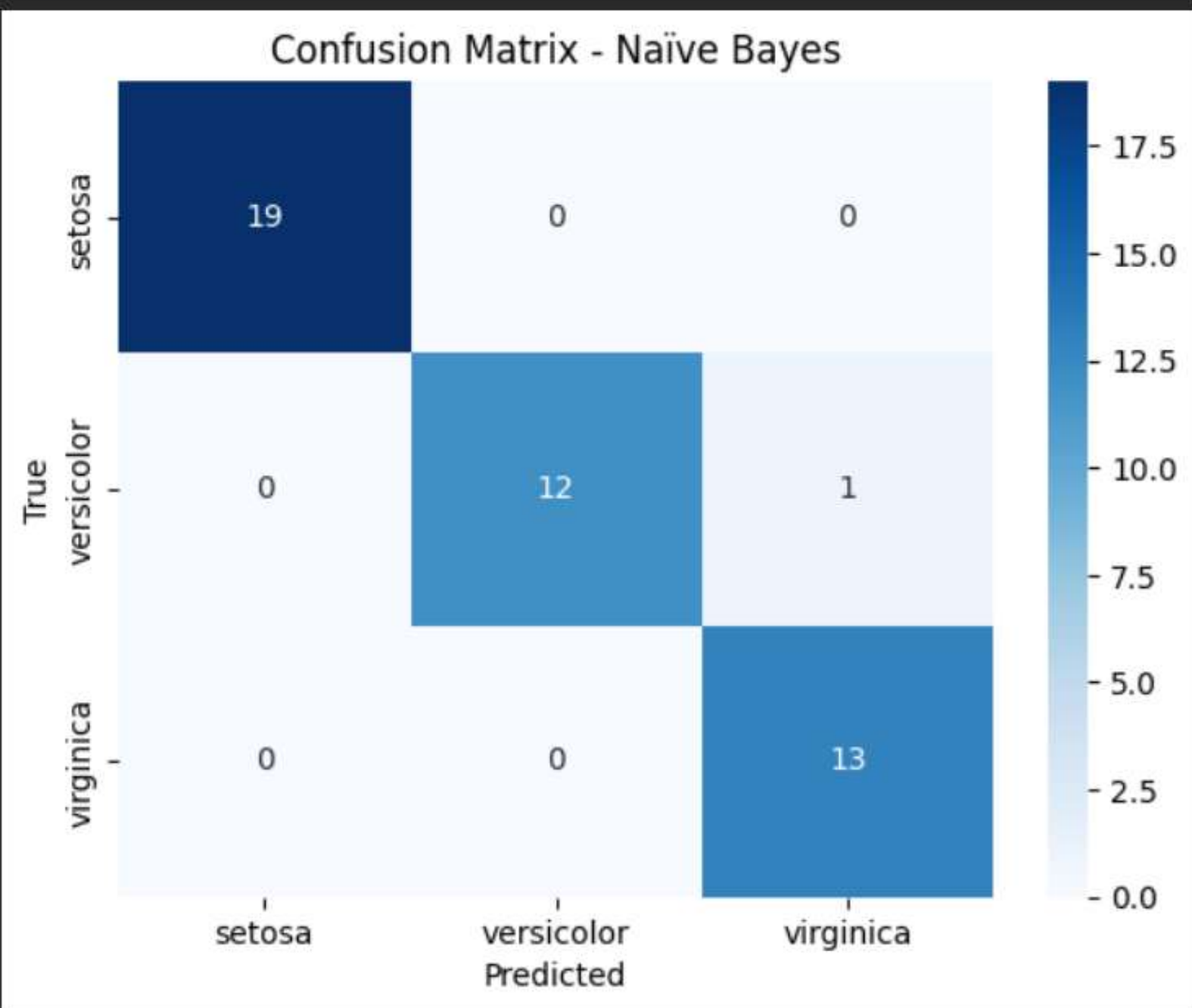




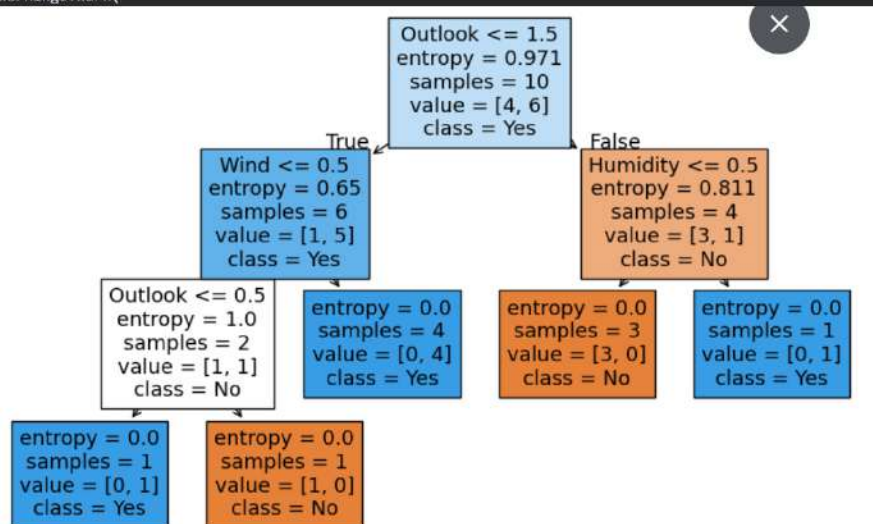
	precision	recall	f1-score	support
0	1.00	0.73	0.85	15
1	0.79	1.00	0.88	15
accuracy			0.87	30
macro avg	0.89	0.87	0.86	30
weighted avg	0.89	0.87	0.86	30

Accuracy Score: 97.78%



Custom K-NN Accuracy: 90.00%

Classified prediction for new sample: Yes
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but DecisionTreeClassifier was fitted with feature names
warnings.warn()



```
Predicted Output after training Backpropagation:  
[[0.15756011]  
 [0.8394638 ]  
 [0.82460241]  
 [0.53282622]]
```

Final Specific Hypothesis: ['Sunny' 'Warm' '?' 'Strong' '?' '?']
Final Generic Hypothesis: [['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?']]

	precision	recall	f1-score	support
Denied	0.31	0.10	0.15	39
Approved	0.60	0.85	0.70	61
accuracy			0.56	100
macro avg	0.45	0.48	0.43	100
weighted avg	0.48	0.56	0.49	100

	precision	recall	f1-score	support
Denied	0.31	0.10	0.15	39
Approved	0.60	0.85	0.70	61
accuracy			0.56	100
macro avg	0.45	0.48	0.43	100
weighted avg	0.48	0.56	0.49	100

Perceptron Prediction Accuracy: 46.67%

Mobile Price Tier Accuracy classification: 25.00%

Multinomial Naive Bayes Accuracy: 86.67%

	Algorithm	Accuracy	F1 Macro Score
0	Logistic Regression	0.973333	0.973679
1	Support Vector Machine	0.986667	0.986870
2	Decision Tree	0.873333	0.869433
3	Random Forest	0.975556	0.976293

House Prediction R2 Score: 0.9412
Root Mean Squared Error: 27.0975

Iris K-NN Accuracy: 100.00%

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	19
versicolor	1.00	1.00	1.00	13
virginica	1.00	1.00	1.00	13
accuracy			1.00	45
macro avg	1.00	1.00	1.00	45
weighted avg	1.00	1.00	1.00	45

```
Initial Specific Hypothesis: ['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']

Step 1 (Instance ['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']): Hypothesis -> ['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']
Step 2 (Instance ['Sunny' 'Warm' 'High' 'Strong' 'Warm' 'Same']): Hypothesis -> ['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']
Step 3 (Instance ['Rainy' 'Cold' 'High' 'Strong' 'Warm' 'Change']): Hypothesis -> ['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']
Step 4 (Instance ['Sunny' 'Warm' 'High' 'Warm' 'Cool' 'Same']): Hypothesis -> ['Sunny' 'Warm' '?' '?' '?' 'Same']
Final Hypothesis: ['Sunny' 'Warm' '?' '?' '?' 'Same']
```



```
Initial Specific Hypothesis: ['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']

Step 1 (Instance ['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']): Hypothesis -> ['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']
Step 2 (Instance ['Sunny' 'Warm' 'High' 'Strong' 'Warm' 'Same']): Hypothesis -> ['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']
Step 3 (Instance ['Rainy' 'Cold' 'High' 'Strong' 'Warm' 'Change']): Hypothesis -> ['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']
Step 4 (Instance ['Sunny' 'Warm' 'High' 'Warm' 'Cool' 'Same']): Hypothesis -> ['Sunny' 'Warm' '?' '?' '?' 'Same']
Final Hypothesis: ['Sunny' 'Warm' '?' '?' '?' 'Same']
```

	precision	recall	f1-score	support
Good	0.28	0.37	0.31	30
Poor	0.18	0.06	0.09	32
Standard	0.47	0.56	0.51	63
accuracy			0.38	125
macro avg	0.31	0.33	0.31	125
weighted avg	0.35	0.38	0.36	125

R2 Score: -0.2817

Mean Absolute Error (MAE): \$8549.66

Expectation-Maximization Cluster Assignments

